RAFAEL DE SANTIS

Full-Stack Software Engineer | AI, Computer Vision & Cloud

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Italian (EU) citizen — authorized to work across the European Union. Open to remote work and global relocation.

SUMMARY

Full-stack software engineer with 4+ years building and operating production systems — distributed services, REST APIs, and React/TypeScript frontends — with deep specialization in computer vision and applied AI. Owns architecture, CI/CD deployment, and production support end to end: from YOLO11 + TensorRT + GStreamer inference at the edge, through Python/FastAPI microservices, to operator dashboards and on-call operations. Track record running systems under 24/7 traffic for JBS and Marfrig (the world's two largest meat processors) and enterprise retail chains. Founded ShopGuard AI as CTO (accelerated by Oracle for Startups, Google for Startups, and Antler), single-handedly owning the entire stack from edge inference to cloud backend — the platform ran across 150 retail stores (4,500+ camera streams) and cut completed thefts by 90% in production environments. Heavy daily use of AI-augmented development tooling (Claude Code, Codex, Cursor) to compress iteration cycles without lowering the quality bar.

CORE SKILLS

Programming Languages: Python, TypeScript, JavaScript (Node.js), Go, C++, SQL

Backend & APIs: FastAPI, Node.js, REST APIs, Microservices, Event-Driven Architecture, WebSockets

Frontend: React, Next.js, TypeScript, Tailwind CSS

Cloud & DevOps: Google Cloud Platform (GCP), Amazon Web Services (AWS), Oracle Cloud (OCI), Docker, CI/CD, PostgreSQL, Supabase, NVIDIA Jetson

AI & LLMs: GPT-4, Claude, LangChain, LangGraph, AutoGen, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Langfuse, Evaluation Pipelines

Computer Vision: YOLO11 / YOLOv8, PyTorch, OpenCV, TensorRT, GStreamer, FFmpeg, MediaPipe, Segment Anything (SAM)

AI-Augmented Development: Claude Code, Codex, Cursor, Agent-Driven Development

PROFESSIONAL EXPERIENCE

Senior AI Engineer III

Jul 2026 – Present

Pix Force — Brazil

Internal move within the Pix Force group, from Promeat AI — returning at senior level to the company where 4 production AI products shipped in 2024. Owner of refinement, performance, and cost optimization for Pix Safety, a production computer vision product.

- Retrained Pix Safety's oversized, under-optimized models into smaller, better-tuned ones — 30% higher accuracy and 45% faster inference in production.
- Cut cloud infrastructure costs by 37% within the first month of the initiative — reserved instances, right-sizing underutilized clusters, and migrating hot-path Python services to C++.
- Mentor junior and mid-level engineers toward independent delivery, and run company-wide training on AI-augmented development (Claude Code, Codex, Lovable) so multiple departments can build their own automations for repetitive processes.

Full-Stack Software Engineer

Jan 2026 – Jun 2026

Promeat AI — Remote, Brazil (part of the Pix Force group)

Production platform serving the world's two largest meat processors — deployed at 4 processing plants (2 JBS, 2 Marfrig). Full-stack ownership across microservices, REST APIs, and React dashboards.

- Live processing of 25,000+ animals counted daily across the deployed footprint — production-critical accuracy with zero manual fallback; the system is the operational source of truth for client plants.
- Restructured ingestion and processing into modular event-driven microservices so high-frequency events flow without bottlenecks or freezes, with clear headroom for further scale.
- Built FastAPI services and React/TypeScript frontends end to end — REST API design, data models, CI/CD deployment, and observability as one coherent system, no handoffs between layers.
- Architected event-driven multi-agent systems (LLM orchestration) pulling live data from client ERP and plant operational systems to trigger autonomous decisions, removing high-frequency manual steps from critical production workflows.
- Delivered a real-time weight-estimation model for a client (vision-based regression over live camera streams) — exceeded the client's 95%+ accuracy requirement, shipping at 98.5% accuracy in real-time prediction.

Founder & CTO
Aug 2025 – Dec 2025

ShopGuard AI — São Paulo, Brazil (Oracle for Startups · Google for Startups · Antler Brasil)

Founded the company and led all engineering. Built an AI-native retail security platform and shipped it into 24/7 production across 150 stores of enterprise retail chains in Brazil — an average of 30 cameras per store, 4,500+ camera streams platform-wide.

- Built real-time edge detection stack — YOLO11 + GStreamer + NVIDIA TensorRT — running at sub-second latency on multi-camera streams (avg. 30 cameras per store) across the 150-store deployed footprint.
- Sole engineer on the entire stack: model training, edge inference, hybrid GCP / Oracle Cloud backend, FastAPI services, React operator dashboards, and client-side integrations.
- Cut completed thefts by 90% in production retail environments — deploying, monitoring, and maintaining the system 24/7, debugging hardware/software issues remotely, and iterating detection accuracy past 80% on real-world failure cases, not benchmarks.
- Worked directly with retail operators on the floor to translate real failures into product changes — shipped weekly based on what actually broke in production.

Co-Founder & Tech Lead
Jan 2024 – Jun 2025

Vision Labs — Brazil

- Shipped multiple computer vision products from pilot to production for enterprise B2B clients — owning architecture, training pipeline, deployment, and ongoing maintenance.
- Integrated GPT-4 and autonomous multi-step workflows (LangChain, AutoGen) on top of detection systems for retail clients — production deployments under real user traffic.
- Drove projects independently from kickoff to live system, including on-call support — acting as the bridge between technical feasibility and customer expectations.

Innovation Projects Lead & Technical Product Owner
Jul 2024 – Feb 2025

Link4Innovation — Brazil

- Led a team of 5+ engineers while staying hands-on in the codebase — pair programming and code review on critical paths alongside sprint management (Scrum / Kanban).
- Owned the project lifecycle end to end for parallel B2B deliveries in industry, logistics, and agribusiness — architecture, backlog refinement, and delivery of working features from ambiguous client requirements.

Full-Stack AI Engineer
Jan 2024 – Jul 2024

Pix Force — Brazil

- Delivered 4 complete AI products to industrial clients — people counting and flow analysis (YOLOv8 + DeepSORT / ByteTrack tracking), employee performance monitoring, vehicle theft detection, and dock dwell-time analytics.
- Owned two of those products end to end (Produtos Mais Fluxo, Vigilare): on-site data collection, custom dataset creation (5,000+ annotated images with data augmentation), model training, dashboards, and deployment — including optimized edge inference on Raspberry Pi and NVIDIA Jetson Nano devices.
- Served as on-site technical liaison for client directors — managed delivery, expectations, and post-sale debugging directly, from code to customer face.

Freelance Software Engineer
Jan 2023 – Dec 2023

Independent — Brazil

- Delivered front-end (React, JavaScript) and back-end (Python, FastAPI, Node.js) projects for early-stage clients — owning architecture, code, and deployment solo, with zero handoffs.
- Built initial computer vision projects (OpenCV, PyTorch) — custom detection and classification pipelines for client-specific automation use cases.

KEY PROJECTS

Industrial Computer Vision Platform — Promeat AI (JBS & Marfrig)
Engineered the ingestion and processing pipeline handling 25,000+ animals counted daily across 4 enterprise processing plants (2 JBS, 2 Marfrig) — modular event-driven microservices with clear headroom for future scale, serving the world's two largest meat processors as their operational source of truth.

Real-Time Edge Detection Pipeline — ShopGuard AI
YOLO11 + TensorRT inference on NVIDIA edge hardware processing parallel RTSP video streams via GStreamer (avg. 30 cameras per store), with cloud-side aggregation on GCP / Oracle Cloud. Built single-handedly and deployed to 150 retail stores — 4,500+ camera streams — under 24/7 production load.

Drone-Based Crowd Density Monitoring — Government Contract
Real-time computer vision system for crowd density estimation at large public events — processing areas with thousands of people simultaneously at a margin of error under 5%. Built under critical latency and reliability constraints and integrated with government event operations infrastructure.

EDUCATION

Software Development & Innovation Analysis
FIAP — Faculdade de Informática e Administração Paulista, São Paulo, Brazil | 2021 – 2022

LANGUAGES

English — Fluent | Portuguese — Native | Spanish — Conversational | Italian — Basic